

Problem Statement- Mr. L does not have an organized area for batteries. At the moment batteries are not being charged and students are unable to use the power tools without them.

1) Who is involved in this problem?

Mr. L, Stem design/Engineering students, IB design and Tech students, parents of students because they are paying for the class and students are not learning as much. The parents who are trying to get their kids an effective education.

2) What are they doing?

The batteries they use for class are taking up space and aren't charged properly. There's also the problem that the batteries are overall in a very messy bunch leading to disorganization and lack of charging the batteries.

3) What unique situations does the problem occur within, if any?

The problem occurs in the lab where the batteries and chargers are stored on a table mashed between a cabinet and drill press. The batteries and chargers are all mixed together in a pile and not all chargers are plugged in. It's happening indoors in a dry room and in the fabrication room which leads to a lot of dust. The batteries are only used during school hours when there is even a class, sometimes used after school. The batteries are in a cramped location making it hard to access the batteries.

4) What is the size of the issue?

A somewhat minor issue that takes the backseat in Mr. L's classroom compared to other problems but still needs to be solved as it's overall slowing down work because of the lack of charged batteries.

5) Where is the problem happening?

In the shop room/work room more specifically on the table in the side of the room next to a drill press and cabinet. In the STEM class of Mr. L in Marshall High School in Virginia.

6) When does the problem occur?

When the batteries are used or just drained because of sitting out. Then the batteries are either not put away or charged batteries are put on the charger or batteries are not put on the chargers at all. It's only really occurring during school hours when demand for the batteries is at its highest and time limits cause students to disregard the organization and just leave the batteries in the corner not charging them. 8:10 to 2:55 mon-fri

7) Why does the problem happen?

Not much space on the table which leads to stacking and no clear battery charging setup. Area is not organized so students may not want to plug batteries back in due

to the mess around the batteries. students also may not know which batteries are charged and will either take drained batteries or plug in fully charged batteries. It all leads to a very messy environment.

8) How often does the problem happen?

It's just a constant state of being but problems only arise when students need to use the batteries which would be everyday and probably worse by the end of the week but if the problem is never alleviated then it's always occurring regardless of when. Throughout the school day and week.

9) How many times did you try to solve this problem with different methods?

There was one attempt at solving the problem but overall no serious measures have been made to fix it. Within this attempt Mr. L tried to keep the batteries organized but didn't solve the integral issue of how the batteries and battery chargers are stored which led to it being disorganized again.